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Display substrate, manufacturing method thereof, and display device

Abstract

A display substrate, a manufacturing method and a display device. In the display substrate, a first power source line includes a transmission portion and a first inlet portion coupled to the transmission portion, the transmission portion is arranged between a display region and the first inlet portion and extends in a first direction, and at least a part of the first inlet portion extends in a second direction. A second power source line includes a peripheral portion and two second inlet portions, the peripheral portion surrounds the display region and is provided with an opening, and two ends of the peripheral portion at the opening are coupled to the two second inlet portions respectively. The first inlet portion is arranged between the two second inlet portions, a first spacing region and a second spacing region are arranged between each second inlet portion and the first inlet portion, the first spacing region has a first distance in the first direction, the second spacing region has a second distance in the first direction, the first spacing region overlaps the encapsulation adhesive region at a first overlapping region, the second spacing region does not overlap the encapsulation adhesive region, and the first distance is greater than the second distance.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/420,050 filed on Jun. 30, 2021, which is the U.S. national phase of PCT Application No. PCT/CN2020/118879 filed on Sep. 29, 2020, which are incorporated herein by reference in their entireties.

TECHNICAL FIELD
(1) The present disclosure relates to the field of display technology, in particular to a display substrate, a manufacturing method thereof, and a display device having the same.
BACKGROUND
(2) In an Organic Light-Emitting Diode (OLED) display device, a metal layer and an organic layer are very sensitive to moisture and oxygen. When the moisture and oxygen enters the display device, the metal layer and the organic layer are oxidized, leading to shrinkage of a light-emitting region and the occurrence of a non-light-emitting part. In addition, the non-light-emitting part caused by the moisture and oxygen expands with the elapse of time, and the aging of elements accelerates, thereby a service life of the display device is shortened remarkably. Currently, in order to protect an interior of the display device from corrosion due to the moisture and oxygen in a better manner, usually the display device is encapsulated in various ways, e.g., using a sealant (e.g., frit glue) and an encapsulation cover plate.
SUMMARY
(3) An object of the present disclosure is to provide a display substrate, a manufacturing method thereof, and a display device having the same, so as to solve the above problem.
(4) In one aspect, the present disclosure provides in some embodiments a display substrate, including a display region and a non-display region surrounding the display region. The non-display region includes an encapsulation adhesive region, and a first power source line and a second power source line are arranged at the non-display region. The first power source line includes a transmission portion and a first inlet portion coupled to the transmission portion, the transmission portion is arranged between the display region and the first inlet portion and extends in a first direction, and at least a part of the first inlet portion extends in a second direction orthogonal to the first direction. The second power source line includes a peripheral portion and two second inlet portions, the peripheral portion surrounds the display region and is provided with an opening, and two ends of the peripheral portion at the opening are coupled to the two second inlet portions respectively. The first inlet portion is arranged between the two second inlet portions, a first spacing region and a second spacing region are arranged between each second inlet portion and the first inlet portion, the first spacing region has a first distance in the first direction, the second spacing region has a second distance in the first direction, the first spacing region overlaps the encapsulation adhesive region at a first overlapping region, the second spacing region does not overlap the encapsulation adhesive region, and the first distance is greater than the second distance.
(5) In some possible embodiments of the present disclosure, the non-display region further includes a fanout region, a plurality of fanout lines is arranged at the fanout region, and the first overlapping region at least partially overlaps the fanout region.

(6) In some possible embodiments of the present disclosure, an orthogonal projection of the first overlapping region onto a base substrate of the display substrate is located within an orthogonal projection of the fanout region onto the base substrate.
(7) In some possible embodiments of the present disclosure, the plurality of fanout lines includes a plurality of target fanout lines, a first portion of each target fanout line is located at the encapsulation adhesive region, an orthogonal projection of the first portion onto the base substrate overlaps an orthogonal projection of the first power source line onto the base substrate and/or an orthogonal projection of the second power source line onto the base substrate, and a distance between orthogonal projections of the first portions of two adjacent target fanout lines onto the base substrate is greater than 1 μm.
(8) In some possible embodiments of the present disclosure, the plurality of fanout lines includes a plurality of target fanout lines, a first portion of each target fanout line is located at the encapsulation adhesive region, an orthogonal projection of the first portion onto the base substrate overlaps an orthogonal projection of the first power source line onto the base substrate and/or an orthogonal projection of the second power source line onto the base substrate, and a distance a between orthogonal projections of the first portions of two adjacent target fanout lines onto the base substrate satisfies a=(0.264˜0.5)*k, where k represents a line width of the target fanout line.
(9) In some possible embodiments of the present disclosure, each second inlet portion includes a first inlet sub-pattern, a second inlet sub-pattern and a third inlet sub-pattern electrically coupled to each other in turn, the first inlet sub-pattern is electrically coupled to a corresponding end of the peripheral portion, the first spacing region is arranged between the first inlet sub-pattern and the first inlet portion, the second spacing region is arranged between the second inlet sub-pattern and the first inlet portion, and the second distance gradually increases in a direction close to the first spacing region until the second distance is equal to the first distance.
(10) In some possible embodiments of the present disclosure, the third inlet sub-pattern extends in the second direction, and a third distance between the third inlet sub-pattern and the first inlet portion in the first direction is equal to a minimum value of the second distance.
(11) In some possible embodiments of the present disclosure, a third spacing region is arranged between the third inlet sub-pattern and the first inlet portion, the display substrate further includes a gate driving circuitry and a plurality of first signal lines configured to provide signals to the gate driving circuitry, and at least a part of each first signal line is arranged at the second spacing region and/or the third spacing region.
(12) In some possible embodiments of the present disclosure, the display substrate further includes a plurality of dummy signal lines, and at least a part of each dummy signal line is arranged at the first overlapping region.
(13) In some possible embodiments of the present disclosure, the display substrate further includes a first gate metal layer, and each dummy signal line is arranged at a same layer, and made of a same material, as the first gate metal layer.
(14) In some possible embodiments of the present disclosure, the display substrate further includes a second gate metal layer, and each dummy signal line is arranged at a same layer, and made of a same material, as the second gate metal layer.
(15) In some possible embodiments of the present disclosure, the display substrate further includes a first gate metal layer and a second gate metal layer, the plurality of dummy signal lines includes a plurality of first dummy signal lines and a plurality of second dummy signal lines, orthogonal projections of the first dummy signal lines onto a base substrate of the display substrate and orthogonal projections of the second dummy signal lines onto the base substrate are arranged alternately, each first dummy signal line is arranged at a same layer, and made of a same material, as the first gate metal layer, and each second dummy signal line is arranged at a same layer, and made of a same material, as the second gate metal layer.
(16) In some possible embodiments of the present disclosure, each dummy signal line extends in

the first direction or the second direction.
(17) In some possible embodiments of the present disclosure, the first overlapping region is covered by the dummy signal lines.
(18) In some possible embodiments of the present disclosure, the display substrate further includes an encapsulation basal layer arranged at the encapsulation adhesive region, surrounding the display region, and broken at a side of the display region adjacent to the first inlet portion.
(19) In some possible embodiments of the present disclosure, the display substrate further includes a first gate metal layer, and the encapsulation basal layer is arranged at a same layer, and made of a same material, as the first gate metal layer.
(20) In some possible embodiments of the present disclosure, the display substrate further includes a base substrate, and a first gate insulation layer, a second gate insulation layer and an interlayer insulation layer laminated one on another on the base substrate in a direction away from the base substrate, the encapsulation basal layer is arranged between the first gate insulation layer and the second gate insulation layer, a plurality of first via-holes is formed in the encapsulation basal layer, the display substrate is further provided with a plurality of groups of first secondary via-holes corresponding to the plurality of first via-holes respectively, each group of first secondary via-holes include a plurality of first secondary via-holes, orthogonal projections of the plurality of first secondary via-holes onto the base substrate are located within an orthogonal projection of a corresponding first via-hole onto the base substrate, and each first secondary via-hole penetrates through the interlayer insulation layer, the second gate insulation layer and the first gate insulation layer.
(21) In another aspect, the present disclosure provides in some embodiments a display device including the above-mentioned display substrate.
(22) In some possible embodiments of the present disclosure, the display device further includes an encapsulation cover plate arranged opposite to the display substrate, and the encapsulation cover plate and the display substrate are sealed through a sealant at an encapsulation adhesive region.
(23) In yet another aspect, the present disclosure provides in some embodiments a method for manufacturing a display substrate. The display substrate includes a display region and a non-display region surrounding the display region, and the non-display region includes an encapsulation adhesive region. The method includes forming a first power source line and a second power source line at the non-display region. The first power source line includes a transmission portion and a first inlet portion coupled to the transmission portion, the transmission portion is arranged between the display region and the first inlet portion and extends in a first direction, and at least a part of the first inlet portion extends in a second direction orthogonal to the first direction. The second power source line includes a peripheral portion and two second inlet portions, the peripheral portion surrounds the display region and is provided with an opening, and two ends of the peripheral portion at the opening are coupled to the two second inlet portions respectively. The first inlet portion is arranged between the two second inlet portions, a first spacing region and a second spacing region are arranged between each second inlet portion and the first inlet portion, the first spacing region has a first distance in the first direction, the second spacing region has a second distance in the first direction, the first spacing region overlaps the encapsulation adhesive region at a first overlapping region, the second spacing region does not overlap the encapsulation adhesive region, and the first distance is greater than the second distance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The following drawings are provided to facilitate the understanding of the present disclosure, and constitute a portion of the description. These drawings and the following embodiments are for

illustrative purposes only, but shall not be construed as limiting the present disclosure. In these drawings,
(2) FIG. **1** is a schematic view showing a display substrate according to one embodiment of the present disclosure;
(3) FIG. **2** is an enlarged view of A**1** in FIG. **1**;
(4) FIG. **3** is another enlarged view of A**1** in FIG. **1**;
(5) FIG. **4** is a schematic view showing a spacing region between a first inlet portion and a second inlet portion according to one embodiment of the present disclosure;
(6) FIG. **5** is another schematic view showing the display substrate according to one embodiment of the present disclosure;
(7) FIG. **6** is an enlarged view of A**2** in FIG. **5**;
(8) FIG. **7** is another enlarged view of A**2** in FIG. **5**;
(9) FIG. **8** is a sectional view of A**2** along B**1**B**2** in FIG. **7**;
(10) FIG. **9** is an enlarged view of A**3** in FIG. **5**;
(11) FIG. **10** is an enlarged view of a portion surrounding a first overlapping region in
(12) FIG. **9**;
(13) FIG. **11** is a sectional view of the portion along C**1**C**2** in FIG. **10**;
(14) FIG. **12** is another sectional view of the portion along C**1**C**2** in FIG. **10**;
(15) FIG. **13** is yet another sectional view of the portion along C**1**C**2** in FIG. **10**;
(16) FIG. **14** is a schematic view showing a situation where short circuit occurs between VDD or VSS and a fanout line;
(17) FIG. **15** is a schematic view showing a situation where a fanout region overlaps a first power source line and a second power source line;
(18) FIG. **16** is an enlarged view of a fanout line at region A**3** in FIG. **15**;
(19) FIG. **17** is yet another schematic view showing the display substrate according to one embodiment of the present disclosure;
(20) FIG. **18** is a schematic view showing the layout of the first power source line and the second power source line in FIG. **17**;
(21) FIG. **19** is an enlarged view of a lower left portion of a frame in FIG. **17**;
(22) FIG. **20** is an enlarged view of A**4** in FIG. **19**;
(23) FIG. **21** is a schematic view showing the second power source line in FIG. **20**;
(24) FIG. **22** is a sectional view of the display substrate along D**1**D**2** in FIG. **17**;
(25) FIG. **23** is a schematic view showing the layout of first via-holes and first secondary via-holes according to one embodiment of the present disclosure; and
(26) FIG. **24** is a schematic view showing the layout of second via-holes according to one embodiment of the present disclosure.

DETAILED DESCRIPTION

(27) In order to make the objects, the technical solutions and the advantages of the present disclosure more apparent, the present disclosure will be described hereinafter in a clear and complete manner in conjunction with the drawings and embodiments.
(28) When encapsulating a display device through frit glue and an encapsulation cover plate, a frit glue layer surrounding the display device is formed at a peripheral region of the display device, then the encapsulation is placed over the display device, and then the frit glue is cured through a laser sintering process, so as to encapsulate functional structures of the display device between a base substrate of the display device and the encapsulation cover plate. However, due to the laser sintering process, the frit glue may shrink, and when it shrinks seriously, an encapsulation effect of an OLED display device may be adversely affected. At this time, it is impossible for the frit glue to effectively prevent the display device from corrosion caused by moisture and oxygen.
(29) As shown in FIGS. **1** and **17**, in a display substrate according to the embodiments of the present disclosure, a first power source line **30** (e.g., a positive power source line VDD) and a

second power source line **40** (e.g., a negative power source line VSS) lead from a pin of a Chip On Film (COF) or an Integrated Circuit (IC) at a side where the COF or IC is arranged, so as to form an inlet region. Due to the limitation of a size of the COF or IC, a distance between VDD and VSS is relatively small, e.g., usually about 50 μm to 100 μm, in the vicinity of the inlet region of the display substrate (e.g., as shown in FIGS. **2** and **6**, a second inlet portion **402** is close to a first inlet portion **302**), and there is no wiring between VDD and VSS, so a surface of each of VDD and VSS away from a base substrate of the display substrate is flat in the vicinity of the inlet region.
(30) In order to save a space of a lower frame of the display substrate (i.e., a frame where the inlet region is located), usually VSS and VDD at the lower frame serve as a frit encapsulation substrate. When the distance between VDD and VSS is relatively small in the vicinity of the inlet region and the surface of each of VDD and VSS away from the base substrate of the display substrate in the vicinity of the inlet region is flat, frit glue between VDD and VSS in the vicinity of the inlet region is adhered to the display substrate at a relatively small effective adhesion width. Hence, when the frit glue is irradiated with laser, the frit glue between VDD and VSS in the vicinity of the inlet region may shrink seriously. At this time, moisture and oxygen may easily enter a display device at this position, so there is a risk of electrochemical corrosion for the display device.
(31) As shown in FIGS. **1**, **3**, **4**, **7**, **8** and **17**, the present disclosure provides in some embodiments a display substrate, which includes a display region **10** and a non-display region surrounding the display region **10**. The non-display region includes an encapsulation adhesive region **20**, and a first power source line **30** and a second power source line **40** are arranged at the non-display region.
(32) The first power source line **30** includes a transmission portion **301** and a first inlet portion **302** coupled to the transmission portion **301**, the transmission portion **301** is arranged between the display region and the first inlet portion **302** and extends in a first direction, and at least a part of the first inlet portion **302** extends in a second direction orthogonal to the first direction.
(33) The second power source line **40** includes a peripheral portion **401** and two second inlet portions **402**, the peripheral portion **401** surrounds the display region and is provided with an opening, and two ends of the peripheral portion **401** at the opening are coupled to the two second inlet portions **402** respectively.
(34) The first inlet portion **302** is arranged between the two second inlet portions **402**, a first spacing region **51** and a second spacing region **52** are arranged between each second inlet portion **402** and the first inlet portion **302**, the first spacing region **51** has a first distance d in the first direction, the second spacing region **52** has a second distance b in the first direction, the first spacing region **51** overlaps the encapsulation adhesive region **20** at a first overlapping region, the second spacing region **52** does not overlap the encapsulation adhesive region **20**, and the first distance d is greater than the second distance b.
(35) To be specific, the display substrate may include the display region and the non-display region. For example, the display region may be of a circular or rectangular shape, and the non-display region may completely surround the display region.
(36) The display substrate may include the first power source line **30** and the second power source line **40**. For example, the first power source line **30** may be a positive power source signal line VDD, and the second power source line **40** may be a negative power source signal line VSS.
(37) The first power source line **30** may include the transmission portion **301** and the first inlet portion **302** electrically coupled to each other. For example, the transmission portion **301** may be arranged between the display region and the first inlet portion **302** and extend in the first direction, and at least a part of the first inlet portion **302** may extend in the second direction. For example, the first direction may be a horizontal direction, and the second direction may be a longitudinal direction.
(38) It should be appreciated that, a shape of the first inlet portion **302** and the quantity thereof will not be particularly defined herein. For example, as shown in FIG. **1**, the first power source line **30** may include two first inlet portions **302** each extending in the second direction and electrically

coupled to the transmission portion **301**. The two first inlet portions **302** may be spaced apart from each other in the first direction, and arranged between the two second inlet portions **402**. For example, as shown in FIG. **5**, the first power source line **30** may include one first inlet portion **302** of a door-like structure, and one side of the first inlet portion **302** close to the transmission portion **301** may be electrically coupled to the transmission portion **301**.
(39) It should be appreciated that, the display substrate may further include a power source pattern arranged at the display region. For example, at least a part of the power source pattern may extend in the second direction, and the power source pattern may be electrically coupled to the transmission portion **301**. For example, the first inlet portion **302** may be electrically coupled to a pin of a binding flexible circuit board in the display substrate. The power source pattern is configured to receive a first power source signal transmitted via the first inlet portion **302** and the transmission portion **301**.
(40) As shown in FIG. **1**, the second power source line **40** may include the peripheral portion **401** and two second inlet portions **402**. The peripheral portion **401** may surround the display region, and electrically coupled to a cathode in the display substrate. At least a part of each second inlet portion **402** may extend in the second direction. For example, each second inlet portion **402** may be electrically coupled to the pin of the binding flexible circuit board in the display substrate.
(41) Taking the display substrate of a quadrilateral shape as an example, the display region may be of a quadrilateral shape, a first side of the display region may defined as a side close to a binding chip, a second side of the display region may be defined as a side opposite to the first side, a third side and a fourth side of the display region may be arranged opposite to each other, the third side may be arranged at a left side of the display region, and the fourth side may be arranged at a right side of the display region.
(42) The peripheral portion **401** may enclose the second side, the third side, the fourth side and a part of the first side of the display region. The opening may be formed in the peripheral portion **401** at the first side, the peripheral portion **401** may be provided with two ends at the opening, and the two ends may be coupled to the two second inlet portions **402** respectively.
(43) The transmission portion **301** may be arranged at the first side of the display substrate and between the display region **10** and the peripheral portion **401**, and the first inlet portion **302** may be arranged between the two second inlet portions **402**. When the first power source line **30** includes one first inlet portion **302**, the first spacing region **51** and the second spacing region **52** may be formed between each second inlet portion **402** and the first inlet portion **302**. When the first power source line **30** includes two first inlet portions **302**, the first spacing region **51** and the second spacing region **52** may be formed between each second inlet portion **402** and an adjacent first inlet portion **302**.
(44) As shown in FIGS. **1**, **3** and **4**, the non-display region may further include the encapsulation adhesive region **20** surrounding the display region, and the frit glue may be formed at the encapsulation adhesive region **20**. The first spacing region **51** may overlap the encapsulation adhesive region **20** at the first overlapping region, and the second spacing region **52** may not overlap the encapsulation adhesive region **20**. At any position in the first spacing region **51**, the first spacing region **51** may have the first distance d in the first direction. For example, the first distance d may be a minimum distance of the first spacing region **51** in the first direction. The second spacing region **52** may have the second distance b in the first direction. For example, the second distance b may be an average distance of the second spacing region **52** in the first direction. For example, the second distance b may be equal to 300 μm, and the first distance d may be greater than the second distance b.
(45) Based on the above-mentioned structure of the display substrate, according to the embodiments of the present disclosure, the first spacing region **51** and the second spacing region **52** may be arranged between the second inlet portion **402** and the first inlet portion **302**, the first spacing region **51** may have the first distance d in the first direction, the second spacing region **52**

may have the second distance b in the first direction, the first spacing region **51** may overlap the encapsulation adhesive region **20** at the first overlapping region, the second spacing region **52** may not overlap the encapsulation adhesive region **20**, and the first distance d may be greater than the second distance b. In this way, it is able to increase a distance between the second inlet portion **402** and the first inlet portion **302** at the first overlapping region in a better manner, and reduce flatness of a surface of the display substrate away from a base substrate at the first overlapping region, thereby to increase an adhesion area and an adhesion strength between the frit glue and the display substrate, ensure an effective adhesion width of the frit glue at the first overlapping region, improve an encapsulation effect, and prevent the occurrence of an encapsulation failure. Hence, according to the display substrate in the embodiments of the present disclosure, it is able to effectively prevent the moisture and oxygen from entering the display device in the vicinity of the first overlapping region without any additional film layer or any improvement in an existing frit encapsulation process, thereby to reduce a risk of electrochemical corrosion for the display device.
(46) As shown in FIGS. **1**, **3**, **15** and **16**, in some embodiments of the present disclosure, the non-display region may further include a fanout region **60**, a plurality of fanout lines **601** may be arranged at the fanout region **60**, and the first overlapping region may at least partially overlap the fanout region **60**.
(47) To be specific, the plurality of fanout lines **601** may be arranged at the fanout region **60**. For example, a plurality of data lines may be arranged at the display region, the plurality of fanout lines **601** may include a plurality of leading wires for the data lines, and the leading wires may be electrically coupled to the data lines respectively.
(48) For example, the plurality of fanout lines may be arranged at a same layer and made of a same material.
(49) For example, the plurality of fanout lines may include a plurality of first fanout lines **6011** and a plurality of second fanout lines **6012**. The plurality of first fanout lines **6011** may be arranged at a same layer and made of a same material, the plurality of second fanout lines **6012** may be arranged at a same layer and made of a same material, and the plurality of first fanout lines **6011** may be arranged at a layer different from the plurality of second fanout lines **6012**. Orthogonal projections of the plurality of first fanout lines **6011** onto the base substrate of the display substrate and orthogonal projections of the plurality of second fanout lines **6012** onto the base substrate may be arranged alternately.
(50) As mentioned above, the first overlapping region may at least partially overlap the fanout region **60**, so as to further reduce the flatness of the surface of the display substrate away from the base substrate at the first overlapping region, thereby to further increase the adhesion area and the adhesion strength between the frit glue and the display substrate. In this way, when the frit glue is irradiated by laser, it is able to effectively reduce a shrinkage level of the frit glue at the first overlapping region and ensure the effective adhesion width of the frit glue, thereby to improve the encapsulation effect and prevent the occurrence of the encapsulation failure.
(51) In some embodiments of the present disclosure, an orthogonal projection of the first overlapping region onto the base substrate of the display substrate may be located within an orthogonal projection of the fanout region **60** onto the base substrate.
(52) Based on the above arrangement mode, it is able to reduce the flatness of the surface of the display substrate away from the base substrate at the first overlapping region to the greatest extent, thereby to further increase the adhesion area and the adhesion strength between the frit glue and the display substrate. In this way, when the frit glue is irradiated by laser, it is able to effectively reduce the shrinkage level of the frit glue at the first overlapping region and ensure the effective adhesion width of the frit glue, thereby to improve the encapsulation effect and prevent the occurrence of the encapsulation failure.
(53) As shown in FIGS. **1**, **15** and **16**, in some embodiments of the present disclosure, the plurality of fanout lines **601** may include a plurality of target fanout lines, a first portion **6010** of each target

fanout line may be located at the encapsulation adhesive region **20**, an orthogonal projection of the first portion **6010** onto the base substrate may overlap an orthogonal projection of the first power source line **30** onto the base substrate and/or an orthogonal projection of the second power source line **40** onto the base substrate, and a distance a between orthogonal projections of the first portions **6010** of two adjacent target fanout lines onto the base substrate may be greater than 1 μm.
(54) For example, the plurality of fanout lines **601** may be arranged at a same layer and made of a same material.
(55) For example, the plurality of fanout lines **601** may include a plurality of first fanout lines **6011** and a plurality of second fanout lines **6012**. The plurality of first fanout lines **6011** may be arranged at a same layer and made of a same material, the plurality of second fanout lines **6012** may be arranged at a same layer and made of a same material, and the plurality of first fanout lines **6011** may be arranged at a layer different from the plurality of second fanout lines **6012**. Orthogonal projections of the plurality of first fanout lines **6011** onto the base substrate and orthogonal projections of the plurality of second fanout lines **6012** onto the base substrate may be arranged alternately.
(56) For example, the distance a between the orthogonal projections of the first portions **6010** of the adjacent target fanout lines onto the base substrate may be 1.5 μm.
(57) As shown in FIG. **14**, when a rigid OLED display substrate is encapsulation using the frit glue and an encapsulation cover plate **97** is attached to the display substrate, the first power source line **30** and the second power source line **40** may easily be pressed into the fanout line **601** (a position as indicated by an arrow in FIG. **14**) due to large granular foreign matters in the frit glue, so a short circuit may occur between each of the first power source line **30** and the second power source line **40** and the fanout line, and thereby an X-line defect may occur.
(58) In the display device according to the embodiments of the present disclosure, the distance between the orthogonal projections of the first portions **6010** of the adjacent target fanout lines onto the base substrate **90** may be greater than 1 μm, so that a distance between the first portions **6010** of the adjacent target fanout lines may increase at the encapsulation adhesive region **20**, and a density of the target fanout lines covered by the first power source line **30** and/or the second power source line **40** at the encapsulation adhesive region **20** may decrease. In this regard, when the encapsulation cover plate **97** is attached to the display substrate, it is able to reduce a probability of the short circuit between each of the first power source line **30** and the second power source line **40** and the fanout line due to the large granular foreign matters.
(59) As shown in FIGS. **1**, **15** and **16**, in some embodiments of the present disclosure, the plurality of fanout lines **601** may include a plurality of target fanout lines, and a first portion **6010** of each target fanout line may be arranged at the encapsulation adhesive region **20**. An orthogonal projection of the first portion **6010** onto the base substrate may overlap an orthogonal projection of the first power source line **30** onto the base substrate and/or an orthogonal projection of the second power source line **40** onto the base substrate, and a distance a between orthogonal projections of the first portions **6010** of two adjacent target fanout lines onto the base substrate may satisfy a=(0.264˜0.5)*k, where k represents a line width of the target fanout line.
(60) To be specific, a may be 0.264 to 0.5 times of k. For example, k may be 3.8 μm.
(61) As shown in FIG. **4**, in some embodiments of the present disclosure, each second inlet portion **402** may include a first inlet sub-pattern **4021**, a second inlet sub-pattern **4022** and a third inlet sub-pattern **4023** electrically coupled to each other in turn, and the first inlet sub-pattern **4021** may be electrically coupled to a corresponding end of the peripheral portion **401**. The first spacing region **51** may be arranged between the first inlet sub-pattern **4021** and the first inlet portion **302**, the second spacing region **52** may be arranged between the second inlet sub-pattern **4022** and the first inlet portion **302**, and the second distance b may gradually increase in a direction close to the first spacing region **51** until the second distance b is equal to the first distance d.
(62) For example, the second distance b may be 30 μm to 290 μm, and the first distance d may be

150 μm to 600 μm.

(63) For example, both the first inlet sub-pattern **4021** and the third inlet sub-pattern **4023** may extend in the second direction, and the second inlet sub-pattern **4022** may extend in a third direction.

(64) For example, the first inlet sub-pattern **4021**, the second inlet sub-pattern **4022**, the third inlet sub-pattern **4023** and the peripheral portion **401** may be formed integrally.

(65) The first inlet sub-pattern **4021** may be electrically coupled to a corresponding end of the peripheral portion **401**, and the third inlet sub-pattern **4023** may be electrically coupled to a driving IC in the display substrate.

(66) When the second distance b gradually increases in the direction close to the first spacing region **51** until it is equal to the first distance d, a distance between the second inlet portion **402** and the first inlet portion **302** may increase gradually in a direction close to the encapsulation adhesive region **20**. In this regard, it is able to ensure the encapsulation performance as well as signal output stability of the second power source line **40**.

(67) As shown in FIG. **4**, in some embodiments of the present disclosure, the third inlet sub-pattern **4023** may extend in the second direction, and a third distance c between the third inlet sub-pattern **4023** and the first inlet portion **302** in the first direction may be equal to a minimum value of the second distance b.

(68) To be specific, the third inlet sub-pattern **4023** may extend in the second direction, the first inlet portion **302** may extend in the second direction, and the third inlet sub-pattern **4023** may be spaced apart from the first inlet portion **302** in the first direction by the third distance c. For example, the third distance may be 80 μm.

(69) The third inlet sub-pattern **4023** and the first inlet portion **302** are both electrically coupled to the driving IC or the flexible circuit board in the display substrate, and the driving IC or the flexible circuit board is of a limited size. Hence, when the third distance c is equal to the minimum value of the second distance b, the distance between the third inlet sub-pattern **4023** and the first inlet portion **302** may be relatively small, so it is able to reduce a difficulty in the electrical coupling of each of the third inlet sub-pattern **4023** and the first inlet portion **302** to the driving IC or the flexible circuit board in the display substrate.

(70) As shown in FIGS. **4** and **19**, in some embodiments of the present disclosure, the second spacing region **52** may be arranged between the second inlet sub-pattern **4022** and the first inlet portion **302**, and a third spacing region **53** may be arranged between the third inlet sub-pattern **4023** and the first inlet portion **302**. The display substrate may further include a gate driving circuitry GOA and a plurality of first signal lines **70** configured to provide signals to the gate driving circuitry GOA, and at least a part of each first signal line **70** may be arranged at the second spacing region **52** and/or the third spacing region **53**.

(71) To be specific, the display substrate may further include the gate driving circuitry GOA. For example, the gate driving circuitry GOA may include shift register units corresponding to gate lines of the display substrate respectively, so as to provide a scanning signal to each gate line.

(72) The display substrate may further include the plurality of first signal lines **70** configured to provide signals to the gate driving circuitry GOA. The plurality of first signal lines **70** may be of various types, e.g., the plurality of first signal lines **70** may include a frame start signal line, a clock signal line, a first level signal line, a second level signal line, etc.

(73) The gate driving circuitry GOA may be arranged at a left side and a right side of the display region, and each first signal line **70** may be electrically coupled to the gate driving circuitry GOA and the driving IC in the display substrate.

(74) As shown in FIGS. **9** and **10**, in some embodiments of the present disclosure, the display substrate may further include a plurality of dummy signal lines **80**, and at least a part of each dummy signal line **80** may be arranged at the first overlapping region.

(75) To be specific, the display substrate may further include the plurality of dummy signal lines **80**

spaced apart from each other, and each dummy signal line **80** may be in a floating state. At least a part of each dummy signal line **80** may be arranged at the first overlapping region.
(76) When at least a part of each dummy signal line is arranged at the first overlapping region, the surface of the display substrate away from the base substrate at the first overlapping region may be uneven, so it is able to further reduce the flatness of the surface of the display substrate away from the base substrate at the first overlapping region, and increase the adhesion area and the adhesion strength between the frit glue and the display substrate. In this way, when the frit glue is irradiated by laser, it is able to effectively reduce the shrinkage level of the frit glue at the first overlapping region and ensure the effective adhesion width of the frit glue, thereby to improve the encapsulation effect and prevent the occurrence of the encapsulation failure.
(77) As shown in FIG. **11**, in some embodiments of the present disclosure, the display substrate may further include a first gate metal layer, and each dummy signal line **80** may be arranged at a same layer, and made of a same material, as the first gate metal layer.
(78) To be specific, the display substrate may include an active layer, a first gate insulation layer **93**, a first gate metal layer, a second gate insulation layer **94**, a second gate metal layer, an interlayer insulation layer **95** and a first source/drain metal layer laminated one on another in a direction away from the base substrate **90**. The active layer may be used to form an active layer pattern of a thin film transistor in the display substrate and some conductive connection patterns. The first gate metal layer may be used to form a gate electrode of the thin film transistor and a gate line of the display substrate. The second gate metal layer may be used to form an electrode plate of a capacitor of the display substrate and an initialization signal line pattern. The first source/drain metal layer may be used to form the first power source line **30**, the second power source line **40**, a data line and some conductive connection members of the display substrate.
(79) When each dummy signal line is arranged at a same layer, and made of a same material, as the first gate metal layer, it is able to form the dummy signal line and the first gate metal layer through a single patterning process, thereby to simplify the manufacture of the display substrate and reduce the manufacture cost.
(80) As shown in FIG. **12**, in some embodiments of the present disclosure, the display substrate may further include a second gate metal layer, and each dummy signal line **80** may be arranged at a same layer, and made of a same material, as the second gate metal layer.
(81) When each dummy signal line **80** is arranged at a same layer, and made of a same material, as the second gate metal layer, it is able to form the dummy signal line and the second gate metal layer through a single patterning process, thereby to simplify the manufacture of the display substrate and reduce the manufacture cost.
(82) As shown in FIG. **13**, in some embodiments of the present disclosure, the display substrate may further include a first gate metal layer and a second gate metal layer. The plurality of dummy signal lines may include a plurality of first dummy signal lines **801** and a plurality of second dummy signal lines **802**, orthogonal projections of the first dummy signal lines **801** onto the base substrate **90** of the display substrate and orthogonal projections of the second dummy signal lines **802** onto the base substrate **90** may be arranged alternately, each first dummy signal line **801** may be arranged at a same layer, and made of a same material, as the first gate metal layer, and each second dummy signal line **802** may be arranged at a same layer, and made of a same material, as the second gate metal layer.
(83) When the first dummy signal line **801** is arranged at a same layer, and made of a same material, as the first gate metal layer and the second dummy signal line **802** is arranged at a same layer, and made of a same material, as the second gate metal layer, it is able to form the first dummy signal line **801** and the first gate metal layer through a single patterning process and form the second dummy signal line **802** and the second gate metal layer through a single patterning process, thereby to simplify the manufacture of the display substrate and reduce the manufacture cost.

(84) In some embodiments of the present disclosure, each dummy signal line **80** may extend in the first direction or the second direction.
(85) To be specific, an extension direction of each dummy signal line may be set according to the practical need, e.g., the dummy signal line may extend in the first direction, the second direction or the third direction, and the third direction may be orthogonal to both the first direction and the second direction.
(86) As shown in FIG. **9**, in some embodiments of the present disclosure, the first overlapping region may be covered by the dummy signal lines **80**.
(87) When the first overlapping region is covered by the dummy signal lines **80**, the surface of the display substrate away from the base substrate **90** at the entire first overlapping region may be uneven, so it is able to reduce the flatness of the surface of the display substrate away from the base substrate **90** at the first overlapping region in a better manner, and increase the adhesion area and the adhesion strength between the frit glue and the display substrate. In this way, when the frit glue is irradiated by laser, it is able to effectively reduce the shrinkage level of the frit glue at the first overlapping region and ensure the effective adhesion width of the frit glue, thereby to improve the encapsulation effect and prevent the occurrence of the encapsulation failure.
(88) As shown in FIGS. **7** and **19**, in some embodiments of the present disclosure, the display substrate may further include an encapsulation basal layer **21** arranged at the encapsulation adhesive region **20**, surrounding the display region, and broken at a side of the display region adjacent to the first inlet portion **302**.
(89) In the display substrate according to the embodiments of the present disclosure, through the encapsulation basal layer **21** at the encapsulation adhesive region **20**, it is able to effectively reflect the laser through the encapsulation basal layer **21** when the frit glue is sintered with the laser, thereby to accelerate a sintering process in a better manner.
(90) In some embodiments of the present disclosure, the display substrate may further include a first gate metal layer, and the encapsulation basal layer **21** may be arranged at a same layer, and made of a same material, as the first gate metal layer.
(91) When the encapsulation basal layer **21** is arranged at a same layer, and made of a same material, as the first gate metal layer, it is able to form the encapsulation basal layer **21** and the first gate metal layer through a single patterning process, thereby to simplify the manufacture of the display substrate and reduce the manufacture cost.
(92) In addition, the first gate metal layer may be made of a metal material having an excellent light-reflecting effect. When the encapsulation basal layer **21** is arranged at a same layer, and made of a same material, as the first gate metal layer, it is able to accelerate the laser sintering process.
(93) As shown in FIGS. **7**, **19**, **20** and **23**, in some embodiments of the present disclosure, the display substrate may further include a base substrate **90**, and a first gate insulation layer **93**, a second gate insulation layer **94** and an interlayer insulation layer **95** laminated one on another on the base substrate **90** in a direction away from the base substrate **90**. The encapsulation basal layer **21** may be arranged between the first gate insulation layer **93** and the second gate insulation layer **94**. A plurality of first via-holes **210** may be formed in the encapsulation basal layer **21**, and the display substrate may be further provided with a plurality of groups of first secondary via-holes **220** corresponding to the plurality of first via-holes **210** respectively. Each group of first secondary via-holes **220** may include a plurality of first secondary via-holes **220**, and orthogonal projections of the plurality of first secondary via-holes **220** onto the base substrate **90** may be located within an orthogonal projection of a corresponding first via-hole **210** onto the base substrate **90**. Each first secondary via-hole **220** may penetrate through the interlayer insulation layer **95**, the second gate insulation layer **94** and the first gate insulation layer **93**.
(94) For example, each of the first via-holes **210** and the first secondary via-holes **220** may be of a rectangular shape. Each first via-hole **210** may have a size of 40 μm*40 μm, and each first secondary via-hole **220** may have a size of 6 μm*6 μm. In one group of first secondary via-holes

220, the plurality of first secondary via-holes **220** may be arranged in an array form, and a distance between two adjacent first secondary via-holes **220** may be 3.5 μm.
(95) The display substrate may further include a barrier layer **91** and a buffer layer **92** arranged between the base substrate **90** and the first gate insulation layer **93**. Before the encapsulation of the display substrate, the buffer layer **92** may be exposed through each first secondary via-hole **220**. In this regard, when the frit glue is applied subsequently, it may permeate into the first secondary via-hole **220** and be in contact with the buffer layer **92**. After the laser sintering, it is able to firmly adhere the encapsulation cover plate **97** to the display substrate.
(96) Moreover, through the plurality of first via-holes **210** and the plurality of groups of first secondary via-holes **220** in the encapsulation basal layer **21**, it is able to increase a roughness level of a surface of the display substrate in contact with a sealant **96**, thereby to improve the encapsulation effect of the display substrate.
(97) The present disclosure further provides in some embodiments a display device including the above-mentioned display substrate.
(98) For example, the display device may be a rigid OLED display device.
(99) For example, the display device may include a rigid OLED watch or bracelet, and an encapsulation for a screen thereof.
(100) According to the display substrate in the embodiments of the present disclosure, the first spacing region **51** and the second spacing region **52** may be arranged between the second inlet portion **402** and the first inlet portion **302**, the first spacing region **51** may have the first distance d in the first direction, the second spacing region **52** may have the second distance b in the first direction, the first spacing region **51** may overlap the encapsulation adhesive region **20** at the first overlapping region, the second spacing region **52** may not overlap the encapsulation adhesive region **20**, and the first distance d may be greater than the second distance b. In this way, it is able to increase a distance between the second inlet portion **402** and the first inlet portion **302** at the first overlapping region in a better manner, and reduce flatness of a surface of the display substrate away from a base substrate at the first overlapping region, thereby to increase an adhesion area and an adhesion strength between the frit glue and the display substrate, ensure an effective adhesion width of the frit glue at the first overlapping region, improve an encapsulation effect, and prevent the occurrence of an encapsulation failure. Hence, according to the display substrate in the embodiments of the present disclosure, it is able to effectively prevent the moisture and oxygen from entering the display device in the vicinity of the first overlapping region without any additional film layer or any improvement in an existing frit encapsulation process, thereby to reduce a risk of electrochemical corrosion for the display device.
(101) When the display device in the embodiments of the present disclosure includes the above-mentioned display substrate, it may also have the above-mentioned beneficial effects, which will not be particularly defined herein.
(102) It should be appreciated that, the display device may be any product or member having a display function, e.g., television, display, digital photo frame, mobile phone or tablet computer.
(103) In some embodiments of the present disclosure, the display device may further include an encapsulation cover plate **97** arranged opposite to the display substrate, and the encapsulation cover plate **97** and the display substrate may be sealed through a sealant **96** at the encapsulation adhesive region **20**.
(104) The present disclosure further provides in some embodiments a method for manufacturing the above-mentioned display substrate. The display substrate includes a display region and a non-display region surrounding the display region, and the non-display region includes an encapsulation adhesive region **20**. The method includes forming a first power source line **30** and a second power source line **40** at the non-display region. The first power source line **30** includes a transmission portion **301** and a first inlet portion **302** coupled to the transmission portion **301**, the transmission portion **301** is arranged between the display region and the first inlet portion **302** and extends in a

first direction, and at least a part of the first inlet portion **302** extends in a second direction orthogonal to the first direction. The second power source line **40** includes a peripheral portion **401** and two second inlet portions **402**, the peripheral portion **401** surrounds the display region and is provided with an opening, and two ends of the peripheral portion **401** at the opening are coupled to the two second inlet portions **402** respectively. The first inlet portion **302** is arranged between the two second inlet portions **402**, a first spacing region **51** and a second spacing region **52** are arranged between each second inlet portion **402** and the first inlet portion **302**, the first spacing region **51** has a first distance d in the first direction, the second spacing region **52** has a second distance b in the first direction, the first spacing region **51** overlaps the encapsulation adhesive region **20** at a first overlapping region, the second spacing region **52** does not overlap the encapsulation adhesive region **20**, and the first distance d is greater than the second distance b.

(105) According to the display substrate manufactured using the method in the embodiments of the present disclosure, the first spacing region **51** and the second spacing region **52** may be arranged between the second inlet portion **402** and the first inlet portion **302**, the first spacing region **51** may have the first distance d in the first direction, the second spacing region **52** may have the second distance b in the first direction, the first spacing region **51** may overlap the encapsulation adhesive region **20** at the first overlapping region, the second spacing region **52** may not overlap the encapsulation adhesive region **20**, and the first distance d may be greater than the second distance b. In this way, it is able to increase a distance between the second inlet portion **402** and the first inlet portion **302** at the first overlapping region in a better manner, and reduce flatness of a surface of the display substrate away from a base substrate at the first overlapping region, thereby to increase an adhesion area and an adhesion strength between the frit glue and the display substrate, ensure an effective adhesion width of the frit glue at the first overlapping region, improve an encapsulation effect, and prevent the occurrence of an encapsulation failure. Hence, according to the display substrate in the embodiments of the present disclosure, it is able to effectively prevent the moisture and oxygen from entering the display device in the vicinity of the first overlapping region without any additional film layer or any improvement in an existing frit encapsulation process, thereby to reduce a risk of electrochemical corrosion for the display device.

(106) As shown in FIGS. **17**, **18**, **19** and **20**, the present disclosure provides in some embodiments a display substrate, which includes a display region **10** and a non-display region surrounding the display region **10**. The non-display region includes an encapsulation adhesive region **20** surrounding the display region. The encapsulation adhesive region **20** includes a corner encapsulation adhesive region **201**, an inlet encapsulation adhesive region **202** and a first encapsulation adhesive region **203** arranged at a first side of the display region **10**. The first encapsulation adhesive region **201** is arranged between the corner encapsulation adhesive region **201** and the inlet encapsulation adhesive region **202**, and an encapsulation basal layer **21** is arranged at the corner encapsulation adhesive region **201**. A second power source line **40**, a gate driving circuitry GOA and a plurality of first signal lines **70** for providing signals to the gate driving circuitry GOA are arranged at the non-display region.

(107) The second power source line **40** includes a power source corner portion **4011**, and a first power source portion **4012** overlapping the first encapsulation adhesive region **203** and extending in a first direction.

(108) A target portion **701** of each first signal line **70** is at least located at the corner encapsulation adhesive region **201** and extends in a second direction orthogonal to the first direction. An orthogonal projection of the target portion **701** onto a base substrate of the display substrate may be located between an orthogonal projection of the encapsulation basal layer **21** onto the base substrate and an orthogonal projection of the first power source portion **4012** onto the base substrate.

(109) To be specific, the display substrate may include the display region **10** and the non-display region. For example, the display region **10** may be of a rectangular shape, and the non-display

region may completely surround the entire display region **10**.
(110) The non-display region may include the encapsulation adhesive region surrounding the display region, and the encapsulation adhesive region may include two corner encapsulation adhesive regions **201**, one inlet encapsulation adhesive region **202** and two first encapsulation adhesive regions **203** which are all arranged at the first side of the display region. The first side may be a side where a lower frame of the display substrate is located. For example, the inlet encapsulation adhesive region **202** may be arranged at a middle position of the lower frame of the display substrate, the two corner encapsulation adhesive regions **201** may be arranged at a lower left corner and a lower right corner of the display substrate respectively, and each first encapsulation adhesive region **203** may be arranged between the inlet encapsulation adhesive region **202** and a corresponding corner encapsulation adhesive region **201**.
(111) It should be appreciated that, the encapsulation adhesive region may further include a second encapsulation adhesive region **204** enclosing a second side, a third side and a fourth side of the display region. The second side may be arranged opposite to the first side, and the third side may be arranged opposite to the fourth side. For example, the third side may be a left side of the display region, and the fourth side may be a right side of the display region.
(112) The display substrate may further include an encapsulation basal layer **21**, at least a part of the encapsulation basal layer **21** may be arranged at the corner encapsulation adhesive region **201**, and the encapsulation basal layer **21** may further include a portion arranged at the second encapsulation adhesive region **204**.
(113) The display substrate may further include the second power source line **40**. For example, the second power source line **40** may include a negative power source signal line VSS. For example, the second power source line **40** may include two power source corner portions **4011** and two first power source portions **4012**. The power source corner portions **4011** may be arranged at the corner encapsulation adhesive regions **201** respectively, and the first power source portions **4012** may be arranged at the first encapsulation adhesive regions **203** respectively. The second power source line **40** may further include a second power source portion **4013** enclosing the second side, the third side and the fourth side of the display region.
(114) For example, each power source corner portion **4011** may overlap the corresponding corner encapsulation adhesive region **201**, or as shown in FIGS. **17**, **20** and **21**, each power source corner portion **4011** may not overlap the corresponding corner encapsulation adhesive region **201**.
(115) As shown in FIG. **1**, for example, the second power source line **40** may include a peripheral portion **401** and two second inlet portions **402**. As shown in FIG. **18**, the peripheral portion **401** may include two power source corner portions **4011**, two first power source portions **4012** and a second power source portion **4013**.
(116) The display substrate may further include a gate driving circuitry GOA. For example, the gate driving circuitry GOA may include shift register units corresponding to gate lines in the display substrate respectively and configured to provide scanning signals to the gate lines. The display substrate may further include the plurality of first signal lines **70** configured to provide signals to the gate driving circuitry GOA. The plurality of first signal lines **70** may be of various types. For example, the plurality of first signal lines **70** may include a frame start signal line, a clock signal line, a first level signal line, a second level signal line, etc. The gate driving circuitry GOA may be arranged at a left side and a right side of the display region, and each first signal line **70** may be electrically coupled to the gate driving circuitry GOA and a driving IC in the display substrate.
(117) The target portion **701** of each first signal line **70** may be at least arranged at the corner encapsulation adhesive region **201** and extend in the second direction. For example, the target portions **701** of the plurality of first signal lines **70** may be spaced apart from each other in the first direction.
(118) Based on the specific structure of the display substrate in the embodiments of the present

disclosure, when the orthogonal projection of the target portion **701** onto the base substrate is located between the orthogonal projection of the encapsulation basal layer **21** onto the base substrate and the orthogonal projection of the first power source portion **4012** onto the base substrate, the transmission basal layer **21**, the target portion **701** of the first signal line **70** and the first power source portion **4012** may together form an encapsulation substrate in contact with a sealant at the corner encapsulation adhesive region **201** and the first encapsulation adhesive region **203**. As a result, it is able to increase a contact area between the sealant and the display substrate and improve the adhesion therebetween, thereby to ensure an excellent encapsulation effect.
(119) In addition, according to the display substrate in the embodiments of the present disclosure, when the transmission basal layer **21**, the target portion **701** of the first signal line **70** and the first power source portion **4012** together form an encapsulation substrate in contact with a sealant at the corner encapsulation adhesive region **201** and the first encapsulation adhesive region **203**, it is able to prevent the formation of a large-size encapsulation basal layer **21** at the corner encapsulation adhesive region **201** and the first encapsulation adhesive region **203**, thereby to reduce the manufacture cost of the display substrate.
(120) As shown in FIG. **21**, in some embodiments of the present disclosure, a boundary of the power source corner portion **4011** away from the display region may be of an arc-like shape, and the power source corner portion **4011** may have a width e in a curvature radius direction. The width e may gradually decrease in a direction from the first encapsulation adhesive region **203** toward the corner encapsulation adhesive region **201**.
(121) To be specific, an orthogonal projection of the power source corner portion **4011** onto the base substrate may partially overlap the orthogonal projections of the plurality of first signal lines **70** onto the base substrate.
(122) When the width e gradually decreases in the direction from the first encapsulation adhesive region **203** toward the corner encapsulation adhesive region **201**, an overlapping area between the second power source line **40** and the target portion **701** of the first signal line **70** may gradually decrease at the corner encapsulation adhesive region **201**, so it is able to reduce a parasitic capacitance generated between the second power source line **40** and the target portion **701** of the first signal line **70** in a better manner, thereby to improve the signal output stability of the first signal line **70**.
(123) In addition, when the target portion **701** of the first signal line **70** is arranged at a same layer, and made of a same material, as a first gate metal layer of the display substrate and the second power source line **40** is arranged at a same layer, and made of a same material, as a first source/drain metal layer of the display substrate, the second power source line **40** may be arranged at a side of the target portion **701** of the first signal line **70** away from the base substrate. When the overlapping area between the second power source line **40** and the target portion **701** of the first signal line **70** gradually decreases at the corner encapsulation adhesive region **201**, it is able to reduce an area of the target portion **701** of the first signal line **70** shielded by the second power source line **40** in a better manner. In addition, because light-reflecting performance of the target portion **701** of the first signal line **70** is superior to that of the second power source line **40**, through the above arrangement mode, it is able to effectively accelerate a laser sintering operation during the encapsulation.
(124) As shown in FIG. **21**, in some embodiments of the present disclosure, a width of the first power source portion **4012** may be greater than a maximum value of the width e in the second direction.
(125) To be specific, the first power source portion **4012** may be reused as the encapsulation substrate and in contact with the sealant, so when the width of the first power source portion **4012** is greater than the maximum value of the width e, it is able to increase a contact area between the first power source portion **4012** and the sealant in a better manner, thereby to improve a laser reflecting effect during the encapsulation and improve the encapsulation efficiency.

(126) In addition, when the width of the first power source portion **4012** is greater than the maximum value of the width e, it is able to increase an overall area of the second power source line **40**, thereby to reduce an IR drop across the second power source line **40**.
(127) In some embodiments of the present disclosure, the orthogonal projection of the encapsulation basal layer **21** onto the base substrate may at least partially overlap an orthogonal projection of the power source corner portion **4011** onto the base substrate.
(128) Through the above arrangement mode, it is able to further reduce the flatness of the surface of the display substrate away from the base substrate at the corner encapsulation adhesive region **201**, thereby to increase adhesion strength between the sealant and the display substrate in a better manner, and improve the encapsulation effect for the display substrate at the corner encapsulation adhesive region **201**.
(129) As shown in FIGS. **17**, **19** and **20**, in some embodiments of the present disclosure, at least a part of the gate driving circuitry GOA may be arranged between the display region and the corner encapsulation adhesive region **201**, and the orthogonal projection of the target portion onto the base substrate of the display substrate may at least partially overlap the orthogonal projection of the power source corner portion **4011** onto the base substrate.
(130) To be specific, at least a part of the gate driving circuitry GOA may be arranged between the display region and the corner encapsulation adhesive region **201**, one end of each first signal line **70** may be electrically coupled to the gate driving circuitry GOA through the corner encapsulation adhesive region **201**, and the other end thereof may be electrically coupled to the driving IC in the display substrate.
(131) The target portion **701** of each first signal line **70** may be arranged at the corner encapsulation adhesive region **201**, and the orthogonal projection of the target portion onto the base substrate of the display substrate may at least partially overlap the orthogonal projection of the power source corner portion **4011** onto the base substrate.
(132) As shown in FIG. **19**, in some embodiments of the present disclosure, the driving IC may be further arranged at the non-display region. Each first signal line **70** may further include a first non-target portion **702** and a second non-target portion **703**, and the first non-target portion **702** may be coupled to the target portion **701** and the second non-target portion **703**. The first non-target portion **702** may be arranged at a side of the first power source portion **4012** away from the display region, at least a part of the first non-target portion **702** may extend in the first direction, and the first non-target portion **702** may be arranged at a same layer, and made of a same material, as the first power source portion **4012**. The second non-target portion **703** may be coupled to the driving IC, and arranged at a layer different from the first non-target portion **702**.
(133) To be specific, each first signal line **70** may include the target portion **701**, the first non-target portion **702** and the second non-target portion **703**. The target portion **701** may be electrically coupled to the gate driving circuitry GOA, the first non-target portion **702** may be coupled to the target portion **701** and the second non-target portion **703**, and the second non-target portion **703** may be coupled to the driving IC.
(134) When the first non-target portion **702** is arranged at a same layer, and made of a same material, as the first power source portion **4012**, it is able to form the first non-target portion **702** and the first power source portion **4012** through a single patterning process, thereby to simplify the manufacture of the display substrate and reduce the manufacture cost.
(135) When the second non-target portion **703** is arranged at a layer different from the first non-target portion **702**, it is able to prevent the occurrence of a short circuit between the second non-target portion **703** and a functional layer of the display substrate arranged at a same layer, and made of a same material, as the first non-target portion **702** (e.g., the first power source line **30** and the second power source line **40**).
(136) In some embodiments of the present disclosure, each of the second non-target portion **703** and the target portion **701** may be arranged at a same layer, and made of a same material, as the

encapsulation basal layer **21**.

(137) Through the above-mentioned arrangement mode, when the target portion **701** is coupled to the gate driving circuitry GOA, no short circuit may occur between the target portion **701** and the second power source line **40**. In addition, the second non-target portion **703** and the target portion may be formed through a single patterning process with the encapsulation basal layer **21**, so as to simplify the manufacture of the display substrate and reduce the manufacture cost.

(138) In some embodiments of the present disclosure, the display substrate may further include a first gate metal layer, and the encapsulation basal layer **21** may be arranged at a same layer, and made of a same material, as the first gate metal layer.

(139) When the encapsulation basal layer **21** is arranged at a same layer, and made of a same material, as the first gate metal layer, it is able to form the encapsulation basal layer **21** and the first gate metal layer through a single patterning process, thereby to simplify the manufacture of the display substrate and the reduce the manufacture cost.

(140) In addition, the first gate metal layer may be made of a metal material having an excellent light-reflecting effect. When the encapsulation basal layer **21** is arranged at a same layer, and made of a same material, as the first gate metal layer, it is able to accelerate the laser sintering process.

(141) In some embodiments of the present disclosure, the display substrate may further include a first gate metal layer and a second gate metal layer. A first target portion of each of a part of the first signal lines **70** may be arranged at a same layer, and made of a same material, as the first gate metal layer, and a second target portion of each of the other part of the first signal lines **70** may be arranged at a same layer, and made of a same material, as the second gate metal layer. Orthogonal projections of the first target portions onto the base substrate of the display substrate and orthogonal projections of the second target portions onto the base substrate may be arranged alternately.

(142) When the first target portion is arranged at a same layer, and made of a same material, as the first gate metal layer, it is able to form the first target portion and the first gate metal layer through a single patterning process, thereby to simplify the manufacture of the display substrate and reduce the manufacture cost.

(143) When the second target portion is arranged at a same layer, and made of a same material, as the second gate metal layer, it is able to form the second target portion and the second gate metal layer through a single patterning process, thereby to simplify the manufacture of the display substrate and reduce the manufacture cost.

(144) When the orthogonal projections of the first target portions onto the base substrate of the display substrate and the orthogonal projections of the second target portions onto the base substrate are arranged alternately, it is able to further increase the roughness level of the surface of the display substrate in contact with the sealant at the corner encapsulation adhesive region **201**, thereby to effectively improve the encapsulation effect of the display substrate.

(145) As shown in FIG. **17**, in some embodiments of the present disclosure, the encapsulation adhesive region may further include a second encapsulation adhesive region **204** enclosing the sides of the display region other than the first side, and the encapsulation basal layer **21** may extend to, and cover, the second encapsulation adhesive region **204**.

(146) To be specific, the second encapsulation adhesive region **204** may enclose the second side, the third side and the fourth side of the display substrate. When the encapsulation basal layer **21** extends to, and covers, the second encapsulation adhesive region **204**, it is able for the encapsulation basal layer **21** at the second encapsulation adhesive region **204** to effectively reflect laser in the case that the frit glue is sintered with the laser, thereby to accelerate the sintering process in a better manner.

(147) As shown in FIGS. **17**, **22** and **23**, in some embodiments of the present disclosure, the display substrate may further include a base substrate, and a first gate insulation layer **93**, a second gate insulation layer **94** and an interlayer insulation layer **95** laminated one on another on the base

substrate in a direction away from the base substrate. The encapsulation basal layer **21** may be arranged between the first gate insulation layer **93** and the second gate insulation layer **94**. A plurality of first via-holes **210** may be formed in the encapsulation basal layer **21**, and the display substrate may be further provided with a plurality of groups of first secondary via-holes **220** corresponding to the plurality of first via-holes **210** respectively. Each group of first secondary via-holes **220** may include a plurality of first secondary via-holes **220**, and orthogonal projections of the plurality of first secondary via-holes **220** onto the base substrate may be located within an orthogonal projection of a corresponding first via-hole **210** onto the base substrate. Each first secondary via-hole **220** may penetrate through the interlayer insulation layer **95**, the second gate insulation layer **94** and the first gate insulation layer **93**.
(148) The display substrate may further include a barrier layer **91** and a buffer layer **92** arranged between the base substrate and the first gate insulation layer **93**. Before the encapsulation of the display substrate, the buffer layer **92** may be exposed through each first secondary via-hole **220**. In this regard, when the frit glue is applied subsequently, it may permeate into the first secondary via-hole **220** and be in contact with the buffer layer **92**. After the laser sintering, it is able to firmly adhere the encapsulation cover plate to the display substrate.
(149) Moreover, through the plurality of first via-holes **210** and the plurality of groups of first secondary via-holes **220** in the encapsulation basal layer **21**, it is able to increase the roughness level of the surface of the display substrate in contact with the sealant, thereby to improve the encapsulation effect of the display substrate.
(150) As shown in FIG. **22**, in some embodiments of the present disclosure, the second power source line **40** may further include a second power source portion **4013** electrically coupled to the power source corner portion **4011** and enclosing the sides of the display region other than the first side. An orthogonal projection of the second power source portion **4013** onto the base substrate may overlap the orthogonal projection of the encapsulation basal layer **21** onto the base substrate, and the second power source portion **4013** may be electrically coupled to the encapsulation basal layer **21** at an overlapping position.
(151) To be specific, the second power source line **40** may further include the second power source portion **4013** enclosing the second side, the third side and the fourth side of the display region, and two ends of the second power source portion **4013** may be electrically coupled to the two power source corner portions **4011** respectively.
(152) For example, at the second side, the third side and the fourth side the orthogonal projection of the second power source portion **4013** onto the base substrate may overlap the orthogonal projection of the encapsulation basal layer **21** onto the base substrate, and the second power source portion **4013** may be electrically coupled to the encapsulation basal layer **21** at an overlapping position at each of the second side, the third side and the fourth side.
(153) When the second power source portion **4013** is electrically coupled to the encapsulation basal layer **21**, it is able to apply a voltage signal at a stable potential to the encapsulation basal layer **21**, thereby to prevent the encapsulation basal layer **21** to be in a floating state. In addition, it is able to reduce a resistance of the second power source line **40**, thereby to effectively reduce an IR drop across the second power source line **40**.
(154) As shown in FIG. **22**, in some embodiments of the present disclosure, the encapsulation basal layer **21** at the second encapsulation adhesive region **204** may include a body portion **230** and a plurality of conductive connection portions **240**. The body portion **230** may enclose the sides of the display region other than the first side. The plurality of conductive connection portions **240** may be electrically coupled to the body portion **230** and spaced apart from each other in an extension direction of the body portion **230**. An orthogonal projection of each conductive connection portion onto the base substrate of the display substrate may overlap the orthogonal projection of the second power source portion **4013** onto the base substrate, and the second power source portion **4013** may be electrically coupled to the corresponding conductive connection portion at each overlapping

position.
(155) According to the display substrate in the embodiments of the present disclosure, it is able to increase the roughness level of the surface of the display substrate in contact with the sealant at an edge of the second encapsulation adhesive region **204**, thereby to improve the encapsulation effect of the display substrate.
(156) In some embodiments of the present disclosure, the body portion **230** and the plurality of conductive connection portions **240** may be formed integrally.
(157) Through the above arrangement mode, it is able to form the body portion **230** and the plurality of conductive connection portions **240** through a single patterning process, thereby to not only ensure the connection performance between the body portion **230** and the plurality of conductive connection portions **240**, but also simplify the manufacture of the display substrate and reduce the manufacture cost.
(158) As shown in FIG. **24**, in some embodiments of the present disclosure, the first power source portion **4012** may be provided with a plurality of second via-holes **4014**.
(159) For example, the plurality of second via-holes **4014** may be arranged in an array form. Each row of second via-holes **4014** in the second direction may include a plurality of second via-holes **4014** spaced apart from each other in the first direction, and two adjacent rows of second via-holes **4014** may be arranged in a staggered manner in the second direction.
(160) For example, an orthogonal projection of each second via-hole **4014** onto the base substrate may be of a rectangular shape, a size of the second via-hole **4014** in the first direction may be 45 μm, and a size of the second via-hole **4014** in the second direction may be 70 μm.
(161) For example, a distance between two adjacent second via-holes **4014** in the first direction may be 45 μm.
(162) In the display substrate, the first power source portion **4012** may be arranged at a side of the interlayer insulation layer **95** away from the base substrate, and the interlayer insulation layer **95** may be exposed through the plurality of second via-holes **4014** in the first power source portion **4012**. In this regard, during the formation of the sealant on the display substrate, the sealant may be in contact with the interlayer insulation layer **95** through the plurality of second via-holes **4014**, so after the laser sintering process, it is able to firmly adhere the encapsulation cover plate to the display substrate. In addition, the first power source portion **4012** may be made of a metal material, so it is able to effectively reflect the laser, thereby to accelerate the sintering process in a better manner.
(163) In some embodiments of the present disclosure, a part of the second inlet portion **402** at the inlet encapsulation adhesive region **202** may be provided with a plurality of second via-holes **4014**.
(164) As shown in FIGS. **1** and **18**, in some embodiments of the present disclosure, a first power source line **30** may be further arranged at the non-display region. The first power source line **30** may include a transmission portion **301** and two first inlet portions **302** coupled to the transmission portion **301**.
(165) The transmission portion **301** may be arranged between the display region **10** and the first power source portion **4012**, and extend in the first direction. The two first inlet portions **302** may be arranged at a side of the transmission portion **301** away from the display region, and extend in the second direction. At least a part of each first inlet portion **302** may be arranged at the inlet encapsulation adhesive region **202**.
(166) For example, the first power source line **30** may include a positive power source signal line VDD.
(167) The first power source line **30** may include the transmission portion **301** and the first inlet portion **302** electrically coupled to each other. The transmission portion **301** may be arranged between the display region and the first inlet portion **302** and extend in the first direction, and at least a part of the first inlet portion **302** may extend in the second direction. For example, the first direction may be a horizontal direction, and the second direction may be a longitudinal direction.

For example, the transmission portion **301** may be arranged at the first side of the display substrate and between the display region **10** and the first power source portion **4012**, and the first inlet portion **302** may be arranged between the two second inlet portions **402**.
(168) For example, the first power source line **30** may include two first inlet portions **302** extending in the second direction and electrically coupled to the transmission portion **301**. The two first inlet portions **302** may be spaced apart from each other in the first direction, and arranged between the two second inlet portions **402**.
(169) A part of the first inlet portion **302** at the inlet encapsulation adhesive region **202**, as an encapsulation substrate, may be in contact with the sealant. Because the first inlet portion **302** is made of a metal material, it is able to effectively reflect the laser, thereby to accelerate the sintering process in a better manner.
(170) As shown in FIG. **19**, in some embodiments of the present disclosure, a part of the first inlet portion **302** at the inlet encapsulation adhesive region **202** may be provided with a plurality of third via-holes **3020**.
(171) For example, the plurality of third via-holes **3020** may be arranged in an array form. Each row of third via-holes **3020** in the second direction may include a plurality of third via-holes **3020** spaced apart from each other in the first direction, and two adjacent rows of third via-holes **3020** may be arranged in a staggered manner in the second direction.
(172) For example, an orthogonal projection of each third via-hole **3020** onto the base substrate may be of a rectangular shape, a size of each third via-hole **3020** in the first direction may be 45 μm, and a size of each third via-hole **302** in the second direction may be 70 μm.
(173) For example, a distance between two adjacent third via-holes **3020** in the first direction may be 45 μm.
(174) In the display substrate, the first inlet portion **302** may be arranged at a side of the interlayer insulation layer **95** away from the base substrate, and the interlayer insulation layer **95** may be exposed through the plurality of third via-holes **3020** in the first inlet portion **302**. In this regard, during the formation of the sealant on the display substrate, the sealant may be in contact with the interlayer insulation layer **95** through the plurality of third via-holes **3020**, so after the laser sintering process, it is able to firmly adhere the encapsulation cover plate to the display substrate.
(175) The present disclosure further provides in some embodiments a display device including the above-mentioned display substrate.
(176) According to the display substrate in the embodiments of the present disclosure, when the orthogonal projection of the target portion onto the base substrate is located between the orthogonal projection of the encapsulation basal layer **21** onto the base substrate and the orthogonal projection of the first power source portion **4012** onto the base substrate, the transmission basal layer **21**, the target portion of the first signal line **70**, the power source corner portion **4011** and the first power source portion **4012** may together form an encapsulation substrate in contact with the sealant at the corner encapsulation adhesive region **201** and the first encapsulation adhesive region **203**. As a result, it is able to increase a contact area between the sealant and the display substrate and improve the adhesion therebetween, thereby to ensure an excellent encapsulation effect.
(177) In addition, according to the display substrate in the embodiments of the present disclosure, when the transmission basal layer **21**, the target portion of the first signal line **70**, the power source corner portion **4011** and the first power source portion **4012** together form the encapsulation substrate in contact with the sealant at the corner encapsulation adhesive region **201** and the first encapsulation adhesive region **203**, it is able to prevent the formation of a large-size encapsulation basal layer **21** at the corner encapsulation adhesive region **201** and the first encapsulation adhesive region **203**, thereby to reduce the manufacture cost of the display substrate.
(178) Hence, when the display device in the embodiments of the present disclosure includes the display substrate, it may also have the above-mentioned beneficial effects, which will not be particularly defined herein.

(179) It should be appreciated that, the display device may be any product or member having a display function, e.g., television, display, digital photo frame, mobile phone or tablet computer.
(180) In some embodiments of the present disclosure, the display device may further include an encapsulation cover plate arranged opposite to the display substrate, and the encapsulation cover plate and the display substrate may be sealed through a sealant at an encapsulation adhesive region.
(181) The present disclosure further provides in some embodiments a method for manufacturing the above-mentioned display substrate. The display substrate includes a display region and a non-display region surrounding the display region, and the non-display region includes an encapsulation adhesive region surrounding the display region. The encapsulation adhesive region includes a corner encapsulation adhesive region **201**, an inlet encapsulation adhesive region **202** and a first encapsulation adhesive region **202** arranged at a first side of the display region, and the first encapsulation adhesive region **202** may be arranged between the corner encapsulation adhesive region **201** and the inlet encapsulation adhesive region **202**. The method includes: forming an encapsulation basal layer **21** at the corner encapsulation adhesive region **201**; and forming a second power source line **40**, a gate driving circuitry GOA and a plurality of first signal lines **70** configured to provide signals to the gate driving circuitry GOA at the non-display region. The second power source line **40** includes a power source corner portion **4011**, and a first power source portion **4012** overlapping the first encapsulation adhesive region **203** and extending in a first direction. A target portion of each first signal line **70** is at least arranged at the corner encapsulation adhesive region **201** and extends in a second direction orthogonal to the first direction. An orthogonal projection of the target portion onto a base substrate of the display substrate is located between an orthogonal projection of the encapsulation basal layer **21** onto the base substrate and an orthogonal projection of the first power source portion **4012** onto the base substrate.
(182) According to the display substrate manufactured using the method in the embodiments of the present disclosure, when the orthogonal projection of the target portion onto the base substrate is located between the orthogonal projection of the encapsulation basal layer **21** onto the base substrate and the orthogonal projection of the first power source portion **4012** onto the base substrate, the transmission basal layer **21**, the target portion of the first signal line **70** and the first power source portion **4012** may together form an encapsulation substrate in contact with a sealant at the corner encapsulation adhesive region **201** and the first encapsulation adhesive region **203**. As a result, it is able to increase a contact area between the sealant and the display substrate and improve the adhesion therebetween, thereby to ensure an excellent encapsulation effect.
(183) In addition, according to the display substrate manufactured using the above-mentioned method, when the transmission basal layer **21**, the target portion of the first signal line **70** and the first power source portion **4012** together form an encapsulation substrate in contact with a sealant at the corner encapsulation adhesive region **201** and the first encapsulation adhesive region **203**, it is able to prevent the formation of a large-size encapsulation basal layer **21** at the corner encapsulation adhesive region **201** and the first encapsulation adhesive region **203**, thereby to reduce the manufacture cost of the display substrate.
(184) It should be appreciated that, the above embodiments have been described in a progressive manner, and the same or similar contents in the embodiments have not been repeated, i.e., each embodiment has merely focused on the difference from the others. Especially, the product embodiments are substantially similar to the method embodiments, and thus have been described in a simple manner.
(185) Unless otherwise defined, any technical or scientific term used herein shall have the common meaning understood by a person of ordinary skills. Such words as “first” and “second” used in the specification and claims are merely used to differentiate different components rather than to represent any order, number or importance. Similarly, such words as “one” or “one of” are merely used to represent the existence of at least one member, rather than to limit the number thereof. Such words as “include” or “including” intends to indicate that an element or object before the word

contains an element or object or equivalents thereof listed after the word, without excluding any other element or object. Such words as "connect/connected to" or "couple/coupled to" may include electrical connection, direct or indirect, rather than to be limited to physical or mechanical connection. Such words as "on", "under", "left" and "right" are merely used to represent relative position relationship, and when an absolute position of the object is changed, the relative position relationship will be changed too.

(186) It should be appreciated that, in the case that such an element as layer, film, region or substrate is arranged "on" or "under" another element, it may be directly arranged "on" or "under" the other element, or an intermediate element may be arranged therebetween.

(187) In the above description, the features, structures, materials or characteristics may be combined in any embodiment or embodiments in an appropriate manner.

(188) The above embodiments are for illustrative purposes only, but the present disclosure is not limited thereto. Obviously, a person skilled in the art may make further modifications and improvements without departing from the spirit of the present disclosure, and these modifications and improvements shall also fall within the scope of the present disclosure.

Claims

1. A display substrate, comprising a display region and a non-display region surrounding the display region, wherein the non-display region comprises an encapsulation adhesive region, and a first power source line and a second power source line are arranged at the non-display region; the first power source line comprises a transmission portion and a first inlet portion coupled to the transmission portion, the transmission portion is arranged between the display region and the first inlet portion and extends in a first direction, and at least a part of the first inlet portion extends in a second direction orthogonal to the first direction; the second power source line comprises a peripheral portion and two second inlet portions, the peripheral portion surrounds the display region and is provided with an opening, and two ends of the peripheral portion at the opening are coupled to the two second inlet portions respectively; and the first inlet portion is arranged between the two second inlet portions, a first spacing region and a second spacing region are arranged between each second inlet portion and the first inlet portion, the first spacing region has a first distance in the first direction, the second spacing region has a second distance in the first direction, the first spacing region overlaps the encapsulation adhesive region at a first overlapping region, the second spacing region does not overlap the encapsulation adhesive region, and the first distance is greater than the second distance; wherein the display substrate further comprises: an encapsulation basal layer arranged at the encapsulation adhesive region, surrounding the display region, and broken at a side of the display region adjacent to the first inlet portion; wherein the display substrate further comprises: a base substrate, and a first gate insulation layer, a second gate insulation layer and an interlayer insulation layer laminated one on another on the base substrate in a direction away from the base substrate, wherein the encapsulation basal layer is arranged between the first gate insulation layer and the second gate insulation layer, a plurality of first via-holes is formed in the encapsulation basal layer, the display substrate is further provided with a plurality of groups of first secondary via-holes corresponding to the plurality of first via-holes respectively, each group of first secondary via-holes comprise a plurality of first secondary via-holes, orthogonal projections of the plurality of first secondary via-holes onto the base substrate are located within an orthogonal projection of a corresponding first via-hole onto the base substrate, and each first secondary via-hole penetrates through the interlayer insulation layer, the second gate insulation layer and the first gate insulation layer.

2. The display substrate according to claim 1, wherein the non-display region further comprises a fanout region, a plurality of fanout lines is arranged at the fanout region, and the first overlapping region at least partially overlaps the fanout region.

3. The display substrate according to claim 2, wherein an orthogonal projection of the first overlapping region onto a base substrate of the display substrate is located within an orthogonal projection of the fanout region onto the base substrate.
4. The display substrate according to claim 3, wherein the plurality of fanout lines comprises a plurality of target fanout lines, a first portion of each target fanout line is located at the encapsulation adhesive region, an orthogonal projection of the first portion onto the base substrate overlaps an orthogonal projection of the first power source line onto the base substrate and/or an orthogonal projection of the second power source line onto the base substrate, and a distance between orthogonal projections of the first portions of two adjacent target fanout lines onto the base substrate is greater than 1 μm.
5. The display substrate according to claim 3, wherein the plurality of fanout lines comprises a plurality of target fanout lines, a first portion of each target fanout line is located at the encapsulation adhesive region, an orthogonal projection of the first portion onto the base substrate overlaps an orthogonal projection of the first power source line onto the base substrate and/or an orthogonal projection of the second power source line onto the base substrate, and a distance a between orthogonal projections of the first portions of two adjacent target fanout lines onto the base substrate satisfies a=(0.264˜0.5)*k, where k represents a line width of the target fanout line.
6. The display substrate according to claim 1, wherein each second inlet portion comprises a first inlet sub-pattern, a second inlet sub-pattern and a third inlet sub-pattern electrically coupled to each other in turn, the first inlet sub-pattern is electrically coupled to a corresponding end of the peripheral portion, the first spacing region is arranged between the first inlet sub-pattern and the first inlet portion, the second spacing region is arranged between the second inlet sub-pattern and the first inlet portion, and the second distance gradually increases in a direction close to the first spacing region until the second distance is equal to the first distance.
7. The display substrate according to claim 6, wherein the third inlet sub-pattern extends in the second direction, and a third distance between the third inlet sub-pattern and the first inlet portion in the first direction is equal to a minimum value of the second distance.
8. The display substrate according to claim 6, wherein a third spacing region is arranged between the third inlet sub-pattern and the first inlet portion, the display substrate further comprises a gate driving circuitry and a plurality of first signal lines configured to provide signals to the gate driving circuitry, and at least a part of each first signal line is arranged at the second spacing region and/or the third spacing region.
9. The display substrate according to claim 1, further comprising a plurality of dummy signal lines, wherein at least a part of each dummy signal line is arranged at the first overlapping region.
10. The display substrate according to claim 9, further comprising a first gate metal layer, wherein each dummy signal line is arranged at a same layer, and made of a same material, as the first gate metal layer.
11. The display substrate according to claim 9, further comprising a second gate metal layer, wherein each dummy signal line is arranged at a same layer, and made of a same material, as the second gate metal layer.
12. The display substrate according to claim 9, further comprising: a first gate metal layer and a second gate metal layer, wherein the plurality of dummy signal lines comprises a plurality of first dummy signal lines and a plurality of second dummy signal lines, orthogonal projections of the first dummy signal lines onto a base substrate of the display substrate and orthogonal projections of the second dummy signal lines onto the base substrate are arranged alternately, each first dummy signal line is arranged at a same layer, and made of a same material, as the first gate metal layer, and each second dummy signal line is arranged at a same layer, and made of a same material, as the second gate metal layer.
13. The display substrate according to claim 9, wherein each dummy signal line extends in the first direction or the second direction.

14. The display substrate according to claim 9, wherein the first overlapping region is covered by the dummy signal lines.
15. The display substrate according to claim 1, further comprising: a first gate metal layer, wherein the encapsulation basal layer is arranged at a same layer, and made of a same material, as the first gate metal layer.
16. A display device, comprising a display substrate, the display substrate comprising a display region and a non-display region surrounding the display region, wherein the non-display region comprises an encapsulation adhesive region, and a first power source line and a second power source line are arranged at the non-display region; the first power source line comprises a transmission portion and a first inlet portion coupled to the transmission portion, the transmission portion is arranged between the display region and the first inlet portion and extends in a first direction, and at least a part of the first inlet portion extends in a second direction orthogonal to the first direction; the second power source line comprises a peripheral portion and two second inlet portions, the peripheral portion surrounds the display region and is provided with an opening, and two ends of the peripheral portion at the opening are coupled to the two second inlet portions respectively; and the first inlet portion is arranged between the two second inlet portions, a first spacing region and a second spacing region are arranged between each second inlet portion and the first inlet portion, the first spacing region has a first distance in the first direction, the second spacing region has a second distance in the first direction, the first spacing region overlaps the encapsulation adhesive region at a first overlapping region, the second spacing region does not overlap the encapsulation adhesive region, and the first distance is greater than the second distance; wherein the display substrate further comprises: an encapsulation basal layer arranged at the encapsulation adhesive region, surrounding the display region, and broken at a side of the display region adjacent to the first inlet portion; wherein the display substrate further comprises: a base substrate, and a first gate insulation layer, a second gate insulation layer and an interlayer insulation layer laminated one on another on the base substrate in a direction away from the base substrate, wherein the encapsulation basal layer is arranged between the first gate insulation layer and the second gate insulation layer, a plurality of first via-holes is formed in the encapsulation basal layer, the display substrate is further provided with a plurality of groups of first secondary via-holes corresponding to the plurality of first via-holes respectively, each group of first secondary via-holes comprise a plurality of first secondary via-holes, orthogonal projections of the plurality of first secondary via-holes onto the base substrate are located within an orthogonal projection of a corresponding first via-hole onto the base substrate, and each first secondary via-hole penetrates through the interlayer insulation layer, the second gate insulation layer and the first gate insulation layer.
17. The display device according to claim 16, further comprising an encapsulation cover plate arranged opposite to the display substrate, wherein the encapsulation cover plate and the display substrate are sealed through a sealant at an encapsulation adhesive region.
18. A method for manufacturing a display substrate, wherein the display substrate comprises a display region and a non-display region surrounding the display region, and the non-display region comprises an encapsulation adhesive region; the method comprises forming a first power source line and a second power source line at the non-display region; the first power source line comprises a transmission portion and a first inlet portion coupled to the transmission portion, the transmission portion is arranged between the display region and the first inlet portion and extends in a first direction, and at least a part of the first inlet portion extends in a second direction orthogonal to the first direction; the second power source line comprises a peripheral portion and two second inlet portions, the peripheral portion surrounds the display region and is provided with an opening, and two ends of the peripheral portion at the opening are coupled to the two second inlet portions respectively; and the first inlet portion is arranged between the two second inlet portions, a first spacing region and a second spacing region are arranged between each second inlet portion and the

first inlet portion, the first spacing region has a first distance in the first direction, the second spacing region has a second distance in the first direction, the first spacing region overlaps the encapsulation adhesive region at a first overlapping region, the second spacing region does not overlap the encapsulation adhesive region, and the first distance is greater than the second distance; wherein the display substrate further comprises: an encapsulation basal layer arranged at the encapsulation adhesive region, surrounding the display region, and broken at a side of the display region adjacent to the first inlet portion; wherein the display substrate further comprises: a base substrate, and a first gate insulation layer, a second gate insulation layer and an interlayer insulation layer laminated one on another on the base substrate in a direction away from the base substrate, wherein the encapsulation basal layer is arranged between the first gate insulation layer and the second gate insulation layer, a plurality of first via-holes is formed in the encapsulation basal layer, the display substrate is further provided with a plurality of groups of first secondary via-holes corresponding to the plurality of first via-holes respectively, each group of first secondary via-holes comprise a plurality of first secondary via-holes, orthogonal projections of the plurality of first secondary via-holes onto the base substrate are located within an orthogonal projection of a corresponding first via-hole onto the base substrate, and each first secondary via-hole penetrates through the interlayer insulation layer, the second gate insulation layer and the first gate insulation layer.